Python, Mathematics & Machine Learning Foundations

WEEK 1

Report for Week 1

1. Overview

Week 1 established the technical base for the internship. It covered core Python programming, the data analysis stack (NumPy, Pandas, Matplotlib), the mathematics behind machine learning, and the first end-to-end supervised learning models built with Scikit-learn. Four tasks were completed, each with a concrete deliverable.

Task	Title	Level	Deliverable
1.1	Python Fundamentals Assignment	Beginner	GitHub repository with solved exercises
1.2	Data Cleaning & EDA	Beginner	Jupyter notebook with cleaned dataset + visualizations
1.3	Math for ML Practice Set	Beginner	PDF/notebook of worked solutions
1.4	Two ML Models with Scikit-learn	Intermediate	Notebook with two trained models + evaluation report

2. Task-wise Details

1.1 Python Fundamentals Assignment

BEGINNER

Task Description

Complete exercises covering variables, loops, functions, OOP, lists, dictionaries, file handling, and exceptions in Python.

Objective

Build fluency in core Python so that later data and ML code can be written cleanly and debugged confidently.

Approach and Methodology

- Solved graded exercises on variables, data types, control flow (if/for/while) and functions with default and keyword arguments.
- Implemented classes with attributes, methods, inheritance and encapsulation to practise object-oriented design.
- Worked with lists, tuples, sets and dictionaries, including comprehensions and nested structures.
- Performed file reading and writing (text/CSV) and wrapped risky operations in try/except/finally blocks with custom error messages.
- Organised all solutions into a GitHub repository with a README, meaningful commits and one file per topic.

Tools and Libraries

Python 3, VS Code / Jupyter, Git and GitHub.

Outcome and Results

All exercises were solved and committed to the repository. *[Add: your actual result, e.g. dataset name, metric values, screenshots reference]*

Key Learnings

- Writing small reusable functions makes later notebooks far easier to maintain.
- Exception handling prevents pipelines from failing silently on bad input.
- Version control gives a clear, reviewable history of progress.

DELIVERABLE GitHub repository with solved exercises.

1.2 Data Cleaning & EDA

BEGINNER

Task Description

Clean a real-world dataset and explore it using NumPy, Pandas, and Matplotlib to uncover patterns and anomalies.

Objective

Convert a raw, messy dataset into an analysis-ready one and extract insights through exploratory data analysis.

Approach and Methodology

- Loaded the dataset and inspected shape, dtypes, summary statistics and missing-value counts.
- Handled missing values (imputation or removal), removed duplicates, fixed inconsistent formats and corrected data types.
- Detected outliers using the IQR rule and box plots, and decided whether to cap, remove or keep them.
- Visualised distributions (histograms), relationships (scatter plots, correlation heatmap) and category comparisons (bar charts).
- Documented findings and anomalies as markdown commentary alongside the code.

Tools and Libraries

NumPy, Pandas, Matplotlib (and Seaborn where used), Jupyter Notebook.

Outcome and Results

A cleaned dataset and a visual EDA were produced, highlighting key patterns and anomalies. *[Add: your actual result, e.g. dataset name, metric values, screenshots reference]*

Key Learnings

- Data quality work usually takes more time than modelling and decides model quality.
- Visual checks reveal problems that summary numbers hide.
- Every cleaning decision should be recorded and justified.

DELIVERABLE Jupyter notebook with cleaned dataset + visualizations.

1.3 Math for ML Practice Set

BEGINNER

Task Description

Solve applied problems in statistics, probability, linear algebra, and basic calculus as used in ML algorithms.

Objective

Understand the mathematical principles that underlie ML algorithms rather than treating them as black boxes.

Approach and Methodology

- Statistics: mean, median, variance, standard deviation, correlation and distributions.
- Probability: conditional probability, Bayes' theorem and independence, linked to classifiers such as Naive Bayes.
- Linear algebra: vectors, matrices, dot products, matrix multiplication, transpose, inverse and eigenvalues, verified with NumPy.
- Calculus: derivatives, partial derivatives and the chain rule, connected to gradient descent and backpropagation.
- Presented worked solutions step by step with short explanations of where each concept is used in ML.

Tools and Libraries

NumPy, Jupyter Notebook, PDF export.

Outcome and Results

The full practice set was solved with worked steps. *[Add: your actual result, e.g. dataset name, metric values, screenshots reference]*

Key Learnings

- Gradients and matrix operations are the core of model training.
- Probability thinking helps interpret model outputs and uncertainty.

DELIVERABLE PDF/notebook of worked solutions.

1.4 Two ML Models with Scikit-learn

INTERMEDIATE

Task Description

Build and evaluate one regression model and one classification model, including preprocessing, train/test split, and metrics.

Objective

Practise the complete supervised learning workflow on both a regression and a classification problem.

Approach and Methodology

- Prepared features: encoding categorical variables, scaling numeric features and splitting data into train and test sets.
- Regression: trained a model (e.g. Linear Regression) and evaluated it with MAE, MSE/RMSE and R-squared.
- Classification: trained a model (e.g. Logistic Regression or Random Forest) and evaluated it with accuracy, precision, recall, F1 and a confusion matrix.
- Used Pipelines to keep preprocessing and modelling reproducible and to avoid data leakage.
- Compared results, interpreted errors and summarised findings in an evaluation report.

Tools and Libraries

Scikit-learn, Pandas, NumPy, Matplotlib.

Outcome and Results

Both models were trained and evaluated. *[Add: your actual result, e.g. dataset name, metric values, screenshots reference]*

Key Learnings

- A proper train/test split is essential for honest evaluation.
- The right metric depends on the problem; accuracy alone can mislead.

- Pipelines make experiments reproducible.

DELIVERABLE Notebook with two trained models + evaluation report.

3. Summary of the Week

- Built a clean Python foundation and a Git-based workflow.
- Completed a full data cleaning and EDA cycle.
- Linked core mathematics to ML algorithms.
- Delivered a working regression and classification model with evaluation.

4. Challenges and How They Were Handled

- Messy data required several cleaning iterations; solved by documenting each step and re-checking distributions.
- Choosing appropriate metrics; solved by matching metrics to the problem type.

5. Outlook

Week 2 moves from classical ML to deep learning with PyTorch, building neural networks, a CNN classifier and structured experiments.